FQE IET 2 Answers 2 - Exams past papers

Radiography and imaging sciences (Kenya Medical Training College)

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KENYA MEDICAL TRAINING COLLEGE
FACULTY OF DIAGNOSTIC SCIENCES
DEPARTMENT OF MEDICAL IMAGING SCIENCES

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN MEDICAL IMAGING SCIENCES

PAPER: IMAGING EQUIPMENT TECHNOLOGY

DATE: 2nd July 2019	**TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER ……

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SECTION I: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The pre-selected temperature of the filament directly controls one of the following
   a. quantity of the electron emitted
   b. the potential difference
   c. the quality of x-ray
   d. the time electrons take to reach the target
2. The x-ray tube target is angled/beveled to provide
   a. a larger actual target to receive electrons
   b. small effective source of x-rays
   c. a larger effective focus
   d. effective source of potential difference
3. Line focus principle describes?
   a. A larger actual focus and large effective focus
   b. A large effective focus that results in a small actual focus
   c. A large effective focus resulting in a small effective focus
   d. None of the above
4. When the anode fails to rotate on pressing 'prep' button, the most probable reason could be?
   a. No kilovolatage peak is available
   b. A break in the starter winding
   c. A break in the supply cable
   d. All of the above
5. The range of kV for mammography is between?
   a. 50-70 kVp
   b. 70-100 kVp
   c. 20-40 kVp
   d. 15-20 kVp
6. Which of the following is true about primary switching;
   a. It happens on the primary circuit where currents and voltage are equal
   b. It happens on both the secondary and primary circuits
   c. It happens on the primary circuit where the voltage is high and the current is low
   d. It happens on the primary circuit where voltage is low and current high
7. Solid state switching in the primary circuit is provided by a pair?
   a. Thyristors
   b. Diodes
   c. Transistors
   d. None of the above
8. Which of the following best describes scatter radiation?
   a. The complete removal of the x-ray by the tissue
   b. The absorbed dose
   c. Primary radiation
   d. Radiation changed in direction as a result of interaction with matter
9. Control of scatter radiation is not done by

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a. Collimation
b. Compression of tissue
c. Increasing kilovoltage
d. Use of high beam diaphragm

10. The main reason for mains voltage compensation in an x-ray machine is to
    a. Produce the right quality of radiograph that matches the selected exposure factors
    b. Save it from damage
    c. Lengthen its working life
    d. Avoid overheating of the filament
11. Which is the most appropriate term for high tension
    a. Ampere
    b. Voltage
    c. Kilovoltage
    d. Electromotive force
12. State the phase difference in a three-phase electricity power generation
    a. 180 degrees
    b. 60 degrees
    c. 90 degrees
    d. 120 degrees
13. In order that a constant potential supply reaches the x-ray tube from high-tension transformer, one of the following sets of components is used
    a. Four diodes, two capacitors, two triode valves
    b. Two diodes, two capacitors
    c. Four diodes, two triode valves
    d. Two diodes, two capacitors, two triode valves
14. The x-ray target is made of?
    a. 10% rhenium 90% tungsten
    b. 90% rhenium 10% tungsten
    c. 50% aluminum and 50% copper
    d. 100% tungsten
15. The x-ray beam dimensions are determined by:
    a. Size of the filament
    b. The degree of electrostatic control applied by the slot containing the filament
    c. Distance between the anode and cathode
    d. All of the above
16. The size of the effective focal spot is controlled by?
    a. The area of the target under bombardment
    b. The angle of the target face
    c. Speed of rotation
    d. A and B
17. The stem of a rotating anode (target) x-ray tube is made of molybdenum because?
    a. Its thermal conducting is poor
    b. Its thermal conducting is better than copper

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c. Its thermal conducting is the same as tungsten
d. Its linear coefficient of expansion is the same as tungsten

18. The ball bearing in a rotating anode is lubricated with which material?
a. Oil
b. Grease
c. Lead and silver
d. Not lubricated at all

19. The x-ray tube housing is internally lined with?
a. Tungsten
b. Lead
c. Molybdenum
d. Aluminium

20. The x-ray tube port is found on the?
a. Glass insert
b. Tube housing
c. Collimator
d. All of the above

21. The following is one of the ways of tube focus selection
a) By having group of mA resistors
b) By varying mA resistors in Autotransformer
c) By varying kVp from autotransformer
d) By having kVp switch

22. Which is the major function of image intensifier in fluoroscopy?
a) Visualize organ motion in real time
b) Visualize insertion of stents in real time
c) Intensify image produced by x-rays
d) Visualize ingested/injected contrast

23. Which is the major function of transducer in an ultrasound machine?
a) Generate image
b) Transmit image and receive image echoes
c) Convert electric power to mechanical power and vise versa
d) Generate electric power

24. What are the functions of fluoroscopic lenses?
a) Focus electrons across the tube from input phosphor to input phosphor
b) Adjust the image produced in output phosphor
c) Focus electrons across the tube from input phosphor to output phosphor
d) Focus off focus electrons

25. All of the following are components of fluoroscopy image intensifier except?
a) Input phosphor

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b) Output phosphor
c) Photocathode
d) Accelerating anode

26. In fluoroscopy the following are components of CCD except
a) Flat panel device
b) Television camera
c) Camera control unit
d) Monitor

27. The maximum field of view which can be obtained with a specific radiographic system is generally limited by?
a) Focal spot size
b) Anode size
c) Anode angle
d) Heel effect

28. A general rule often employed for determining the kilovoltage to be used when X-raying is:
a) The kilovoltage should be as high as other factors will permit
b) The kilovoltage should be as low as other factors will permit
c) The kilovoltage is always a fixed value and cannot be changed
d) The kilovoltage is not an important variable and can be changed over a wide range without affecting the radiograph

29. The voltage and waveform applied to the X-ray tube by a high-voltage transformer primarily determines the:
a) Quantity of radiation
b) Duration of exposure
c) Penetrating ability
d) X-ray beam divergence

30. Collimators are used to:
a) Filter the radiation beam
b) Reduce the radiation beam spread
c) Increase film latitude
d) Decrease film latitude

31. The target of an X-ray tube is often made out of tungsten because?
a) It has a high atomic mass which will result in more X-rays being generated due to atomic particle interactions

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b) It is an inexpensive material that is easy to machine

c) It have very high thermal conductivity which makes it easy to cool

d) None of the above

32. The filament of an X-ray tube is often made out of tungsten because?
   a) It has a high atomic mass which will result in more X-rays being generated due to atomic particle interactions

   b) It is an inexpensive material that is easy to machine

   c) It has very high thermal conductivity which makes it easy to boil off electrons for emission

   d) It has a high atomic number and high melting point

33. The image intensifier's input phosphor is generally composed of:
   a) cesium iodide
   b) zinc cadmium sulfide
   c) gadolinium oxysulfide
   d) calcium tungstate

34. Accurate operation of the AEC device is dependent on:
   1. Thickness and density of the object
   2. Positioning of the object with respect to the ionization chamber
   3. Beam restriction

   a) 1 only
   b) 1 and 2 only
   c) 2 and 3 only
   d) 1, 2, and 3

35. A radiograph made with a parallel grid demonstrates decreased density on its lateral edges. This is most likely caused by:

   a) static electrical discharge
   b) improper tube angle
   c) decreased SID
   d) the grid off-centered

36. Grid ratio is defined as the relationship between the height of the lead strip and the:
   a) width of the lead strip
   b) distance between the lead strips
   c) number of lead strips per inch
   d) angle of the lead strip

37. Examples of primary radiation barriers include:
(1). x-ray room walls (2). control booth (3). lead aprons

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a) 1 only
b) 1, 2, and 3
c) 1 and 2 only
d) 2 and 3 only

38. If the primary coil of the high-voltage transformer is supplied by 220 V and has 150 turns, and the secondary coil has 75,000 turns; what is the voltage induced in the secondary coil?

a) 110 V
b) 75 kV
c) 110 kV
d) 75 V

39. Which of the following circuit devices operate(s) on the principle of self-induction?
(1). Autotransformer (2). Choke coil (3). high-voltage transformer

a) 1 only
b) 1 and 2 only
c) 2 and 3 only
d) 1, 2, and 3

40. Which of the following statement(s) regarding transformer laws is (are) correct?

1. The voltage and current values are increased with a step-up transformer
2. The voltage is directly related to the number of turns in the two coils
3. The product of voltage and current in the two circuits must be equal

a) 1 only
b) 1 and 2 only
c) 2 and 3 only
d) 1, 2, and 3

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. List five components of a light beam diaphragm **(5 marks)**
42. Explain 'stroboscopic' effect **(5 marks)**
43. List six factors that will extend x-ray tube life **(5 marks)**
44. List 5 components of mammography equipment **(5 marks)**
45. State 5 components of ultrasound equipment **(5 marks)**
46. State the importance of incorporating safety devices in imaging equipment **(5 marks)**
47. Explain the terms; primary and secondary exposure switching **(5 marks)**
48. State the changes that occur in the mains voltage **(5 marks)**

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SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. Discuss filament circuit

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